

Agent96: A Utility-Safe Hybrid Negotiator for the HAN 2026 Competition

Team 406

Human-Agent Negotiation League - ANAC 2026

Abstract. Agent96 is a hybrid negotiator for the HAN 2026 Human-Agent Negotiation league. Its main negotiation decisions are made by deterministic code: it enumerates valid outcomes, protects reservation value, models the opponent from observed offers, and selects offers using an adaptive concession strategy. The language-model layer is intentionally bounded. In the submitted configuration, Qwen 3 4B-Instruct is used only when interaction looks human-like, mainly for short natural-language messages and soft social hints; it is not trusted with final offer validity, acceptance, or utility arithmetic. This design lets Agent96 behave as a steady, utility-aware negotiator while still communicating in a cooperative human-facing style.

1. Competition Setting and Design Goal

HAN 2026 uses the alternating offers protocol with structured multi-issue offers and optional text communication. A successful agent must therefore solve two problems at the same time: it must make valid utility-safe bargaining decisions, and it must communicate those decisions in a way that is understandable to a human partner. Agent96 was designed around a simple rule: deterministic code negotiates, and the LLM communicates.

This separation is important because the formal offer determines the final utility. A small local LLM can produce persuasive language, but it can also hallucinate package details, misread utility values, or suggest invalid outcomes. Agent96 avoids those risks by computing all binding decisions from the actual outcome space and utility function. The LLM can help phrase or interpret, but it cannot accept below the protected floor or invent an agreement.

2. Turn-Level Behavior

On every negotiation turn, Agent96 follows a fixed pipeline:

- 1 Read the current structured offer and any attached text.
- 2 Update the history of opponent offers and issue-value frequencies.
- 3 Refresh an estimated opponent utility model for scoring and diagnostics.
- 4 Optionally read human-like text as a social signal.
- 5 Compute a target utility from time, reservation value, and opponent behavior.
- 6 Accept only if the offer clears the current target, planned next offer, or late protected floor.
- 7 Otherwise, score valid counteroffers and return the best utility-safe package.
- 8 Attach a short safe message, either from templates or the cached LLM message bank.

The agent can run without Ollama and still negotiates normally. If LLM calls fail or time out, it falls back to deterministic messages and keeps the same negotiation strategy.

3. Deterministic Negotiation Core

3.1 Outcome Search and Utility Floors

Agent96 enumerates the complete discrete outcome space exposed by NegMAS, evaluates every outcome with its own utility function, and sorts outcomes from best to worst. This ranked list is the basis for opening offers, concessions, acceptance checks, and fallback behavior.

The first safety rule is reservation protection: offers at or below reserved value are rejected. Above that, the agent compares incoming offers against a time-dependent target and against the utility of the counteroffer it was about to make. Near the deadline, it can accept offers above a protected floor to avoid unnecessary no-deals.

3.2 Adaptive Concession

The target utility follows a Boulware-style concession curve:

$$\text{target}(t) = \text{best} - t^e * (\text{best} - \text{floor})$$

where t is relative time, best is the best available utility, and floor is the current protected value above reservation. The exponent is adjusted from recent opponent behavior: if opponent offers improve for Agent96, it stays firmer; if offers stagnate or harden, it concedes faster. This creates a tough early style, a broader middle search, and a practical closing phase.

Phase	Relative time	Main behavior
Early	< 0.62	High utility target, limited concession
Mid	0.62-0.84	More opponent-aware candidate scoring
Closing	0.84-0.96	Increase agreement pressure while protecting utility
Panic	≥ 0.96	Search for safe closing deals above reservation

Table 1: Current phase schedule used by the submitted Agent96 class.

3.3 Offer Scoring

Counteroffers are selected from outcomes above the current target or floor. Candidate scoring combines normalized own utility, estimated opponent acceptability, similarity to recent opponent offers, domain-specific scoring, small offer variety, and preference-concealment signals:

$$\text{score}(o) = w_{\text{own}} U(o) + w_{\text{opp}} A(o) + w_{\text{fit}} F(o) + w_{\text{dom}} D(o) + \text{variety} + \text{concealment}$$

The weights shift as time advances. Early negotiation emphasizes own utility. Later phases give more weight to opponent acceptability and similarity to recent opponent offers so the agent can close agreements rather than repeatedly proposing unreachable packages.

4. Opponent Modeling

Agent96 builds a lightweight opponent model from observed offers. For each issue, it tracks how often the opponent proposes each value, with slight decay so recent offers matter more. Issues where the opponent is consistent have lower entropy and are treated as more important. Values repeatedly requested by the opponent, or numerically close to those values, receive higher estimated opponent utility.

The resulting model is exposed as a `MappingUtilityFunction`. This helps the agent choose offers that are plausible for the other side and also aligns with HAN's interest in preference modeling. The model is intentionally simple and robust: it does not require training data, external services, or long histories.

5. Domain Awareness and Closing Rules

The agent uses general domain cues from issue names and value shapes. In trade-like domains, it scores quantity-price packages using own utility, opponent acceptability, and smoothness relative to recent offers. In allocation-like domains such as item division, it balances own utility with the opponent's repeated value preferences while trying not to reveal its full ranking too directly.

Near the deadline, Agent96 also watches for repeated-anchor opponents that keep offering the same package. If the opponent appears to have a hard threshold and there is still a positive deal available, the agent searches for a high-anchor-fit package that remains above reservation. This is a narrow rescue mechanism for otherwise avoidable no-deals, not the main strategy.

6. Communication Layer

The submitted class defaults to `use_llm_text="auto"`, `use_llm_social="auto"`, and `use_llm_reasoning=False`. This means Agent96 does not rely on LLM decisions during ordinary bot negotiations. It uses LLM text only when the interaction looks human-like, or when human mode is explicitly enabled.

Generated messages are intentionally short and generic. They refer to "this package", "movement", "balance", and "agreement" rather than exact issue values. This avoids a common failure mode in which an LLM claims a specific package change that the formal offer did not actually make. Messages are filtered to avoid revealing reserved value, utility, issue rankings, or other private preference information.

For latency control, Agent96 caches message variants by intent and negotiation phase. If the LLM is unavailable, the agent marks it unavailable and uses deterministic fallback text. This keeps negotiation behavior stable in both local and server environments.

7. Evaluation Summary

Agent96 was evaluated locally against the provided non-LLM opponents, template-based negotiators, generated scenarios, and stochastic human-proxy opponents. The most important checks were agreement rate, normalized advantage, raw advantage, opponent-model share, step count, runtime, and LLM call count.

Evaluation slice	Agreement	Mean normalized advantage	Notes
TemplateBased + BOANeg, generated 20 seeds	1.0000	0.6753	No regression after final patch
TemplateBased + BOANeg, static+generated 10 seeds	0.9167	0.5894	No regression after final patch
Human proxies, static+generated 5 seeds	1.0000	0.7234	No regression after final patch
Hard-threshold generated opponent, 20 seeds	0.8583	0.2380	Improved no-deal handling

Table 2: Focused local validation results for the final submission agent.

These local results are not official leaderboard guarantees, but they were useful for checking that final changes improved no-deal handling while preserving performance against stronger time-based, BOA, template, and human-proxy opponents.

8. Safety and Reliability

The final submission prioritizes reliability over clever but fragile reasoning. The model name is locked to `qwen3:4b-instruct` for competition compatibility. The stable class does not ask the LLM to select final offers. It can compile, instantiate, and negotiate without Ollama. Validation before submission included Python compilation, the focused unit test suite, a HAN CLI smoke test, and a clean extracted-zip import check.

9. Conclusion

Agent96 is a practical hybrid agent: deterministic where correctness matters, language-enabled where human interaction matters. It searches the complete outcome space, protects reservation value, adapts concession pressure over time, models the opponent from observed offers, and communicates with safe concise text. This architecture is intended to perform consistently across automated local opponents while remaining suitable for human-facing HAN negotiations.

References

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